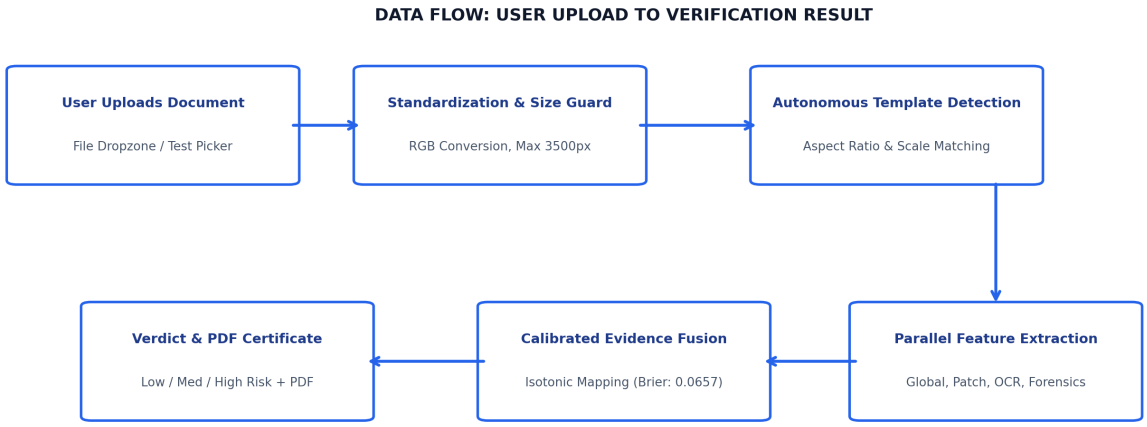
DocVerify — Phase 4: Multi-Modal Prototype Exploration

Official Technical Milestone Report & Engineering Specification (Phase 4)

1. Phase Executive Summary & Objectives

Phase 4 explored combining optical character recognition (EasyOCR), field-level bounding box proposals (YOLO), and digital image forensics. It proved that a multi-modal defense is required, but demonstrated that naive score averaging produced unstable, unpredictable classification verdicts.

2. Architectural Diagram & Pipeline Workflow



3. Technical Specifications & Milestone Deliverables

Dimension / Metric	Implementation / Specification	Significance & Impact
OCR Exploration	EasyOCR Text Extraction + Regex Syntax Rules	Extracted dates and numbers to catch syntactic forgeries (e.g. impossible dates like 32/13/1985).
YOLO Investigation	YOLOv5/v8 Object Detection Proposals	Tested bounding box proposals; identified high boundary jitter and severe compute overhead.
Forensic Exploration	Error Level Analysis (ELA) + Sensor Noise	Identified that JPEG re-compression and high-frequency noise expose copy-pasted text/photos.
Fusion Strategy	Experimental Naive Score Averaging	Equal weighting caused noisy signals to overturn confident visual predictions, proving the need for statistical fusion.

4. Key Discoveries, Challenges & Next-Phase Directives

- Confirmed that no single modality is sufficient on its own (visual, text, and forensics must collaborate).
- Discovered that naive score averaging is fatally flawed; established requirement for trained logistic fusion.
- Observed that YOLO bounding boxes cut through text lines, creating downstream OCR recognition errors.

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